SalaryScope

Model Analytics Report

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Regression Model: Random Forest

Test R^2: 0.9639

Cross-Validation R^2 (Mean): 0.9621

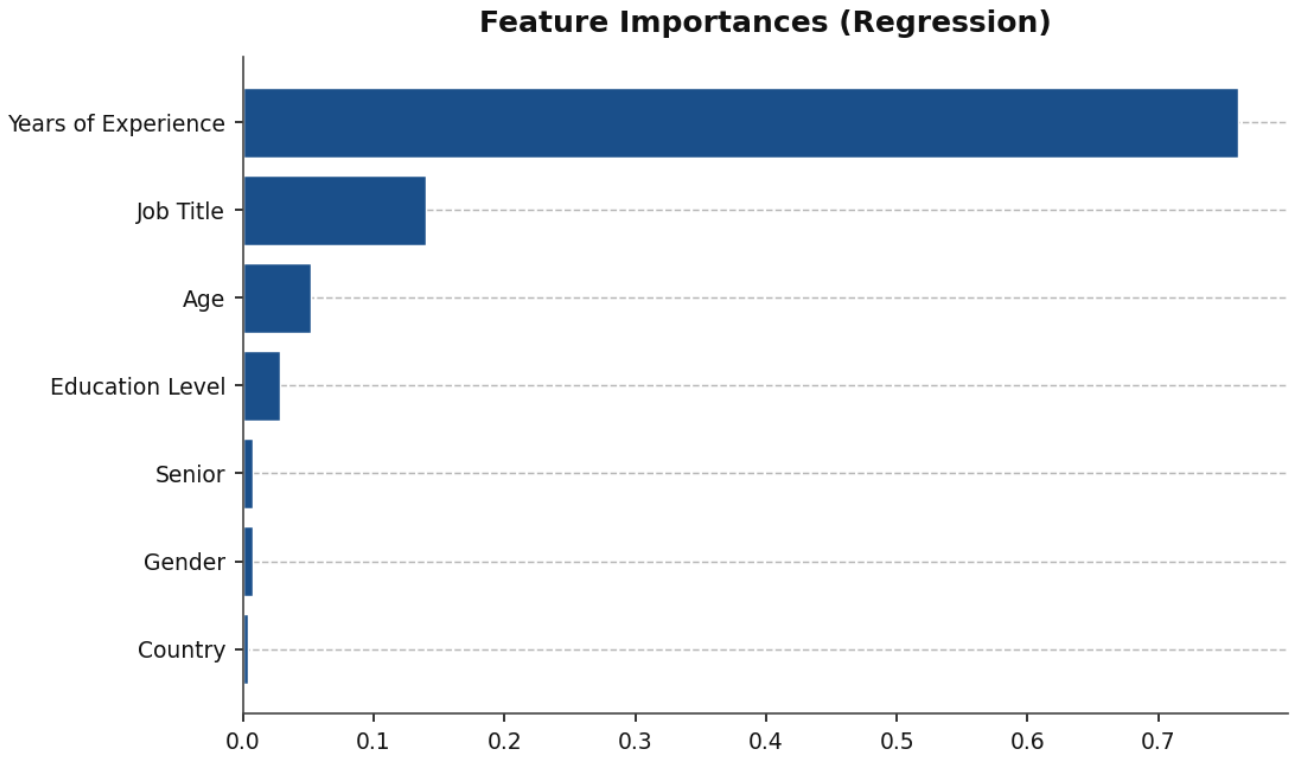
MAE: $4,926.80

RMSE: $9,760.51

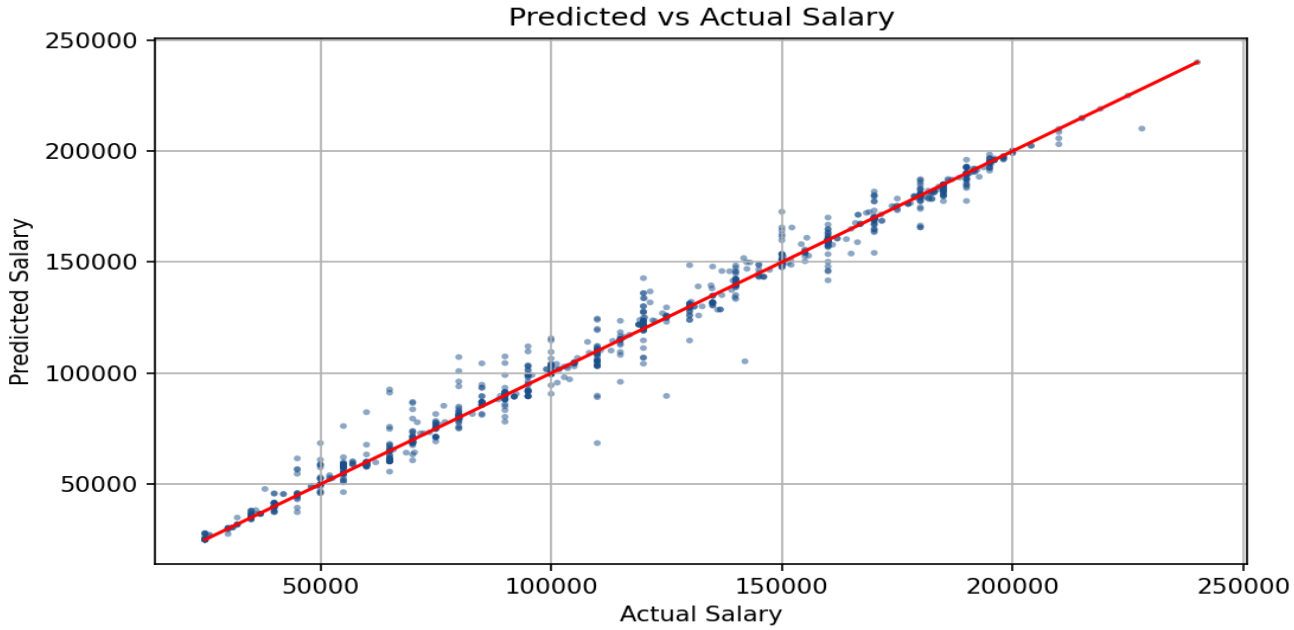
Model Comparison (Regression)

Model	R^2	MAE	RMSE
Random Forest (GridSearchCV)	0.9639	$4,927	$9,761
XGBoost (GridSearchCV)	0.9595	$5,862	$10,337
Gradient Boosting Regression	0.8922	$12,405	$16,871
Decision Tree Regression	0.8622	$13,974	$19,080
Linear Regression	0.7996	$16,884	$23,009

Feature Importance — Regression Model

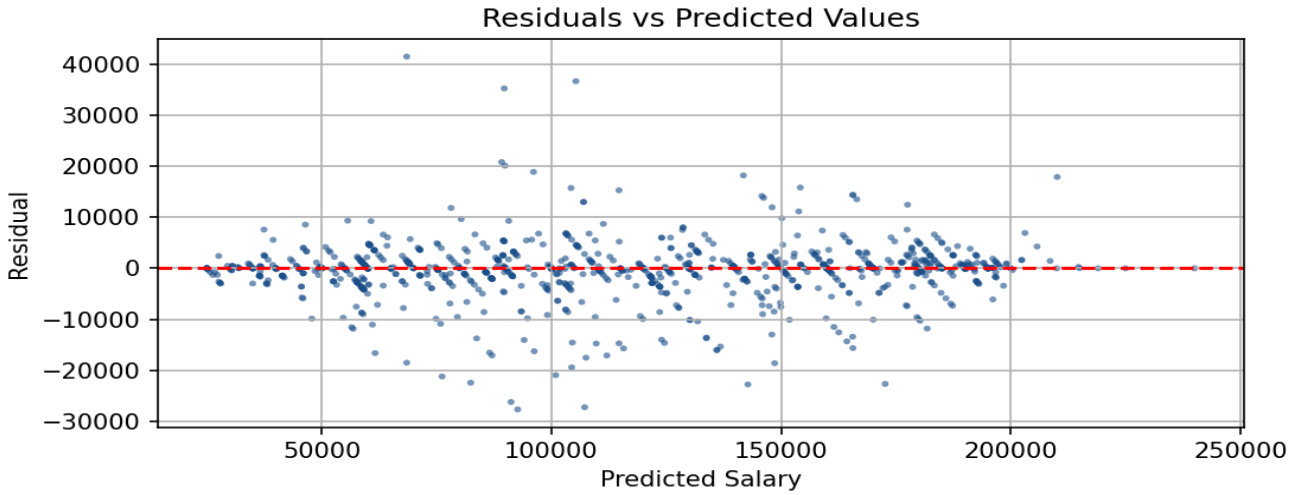


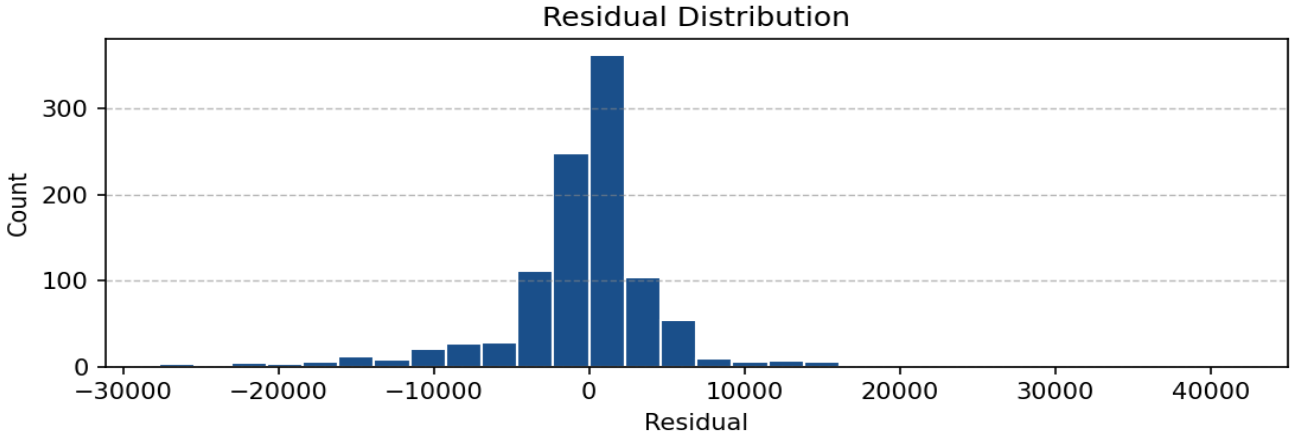
Predicted vs Actual Analysis



Interpretation: Points closer to the diagonal line represent accurate predictions. Greater dispersion indicates prediction error.

Residual Analysis





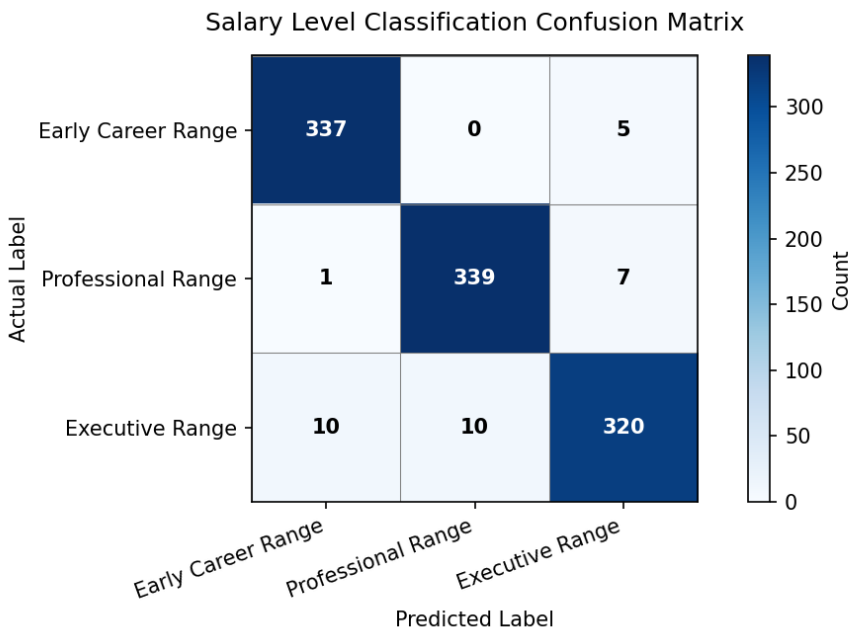
Interpretation: Residuals centered around zero indicate balanced model errors. A symmetric distribution suggests stable predictive behavior.

Salary Level Classification Model

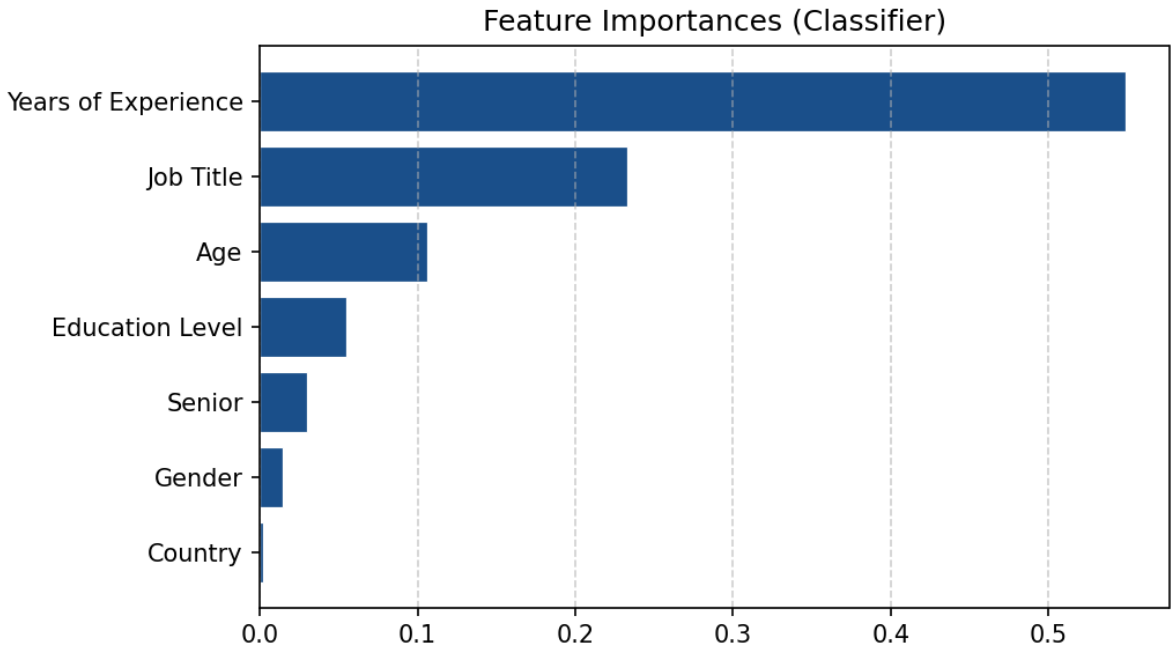
Performance Metrics

Accuracy: 0.9679
Precision (Macro): 0.9679
Recall (Macro): 0.9678
F1 Score (Macro): 0.9678

Confusion Matrix



Feature Importance (Classifier)



Generated by SalaryScope

Career Stage Clustering Model

This model segments individuals into career stages using KMeans clustering based on Years of Experience and Education Level. A derived Career Score feature enhances separation between progression levels.

Clustering Quality Metrics

Silhouette Score: 0.4672

Davies-Bouldin Score: 0.8306

Model Configuration

Model Type: KMeans Career Clustering Pipeline

Training Dataset: Salary.csv

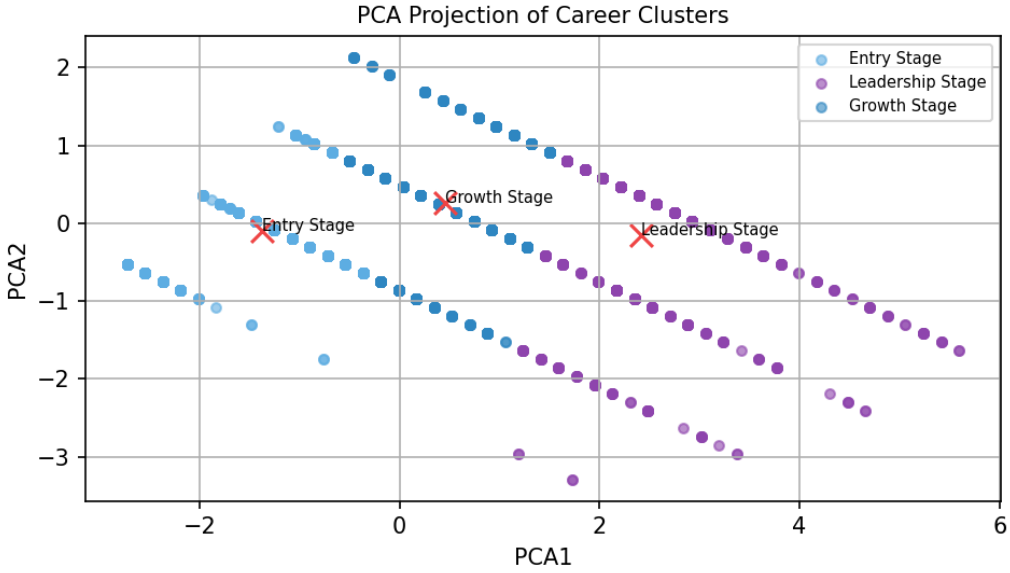
Dataset Shape: (5144, 11)

Features Used: Years of Experience, Education Level

Engineered Feature: Career Score = Experience + (Education Level * 2)

Number of Clusters: 3

Cluster Visualization (PCA Projection)



Note: The structured patterns in this PCA visualization reflect the linear relationships between experience, education, and the derived career score used for clustering.

Interpretation: The clustering model groups individuals into career stages based on experience and education. The PCA visualization shows separation between clusters, while the derived career score enhances progression-based segmentation.

Association Rule Mining (Apriori)

Rule Metrics Summary

Total Rules: 62

Max Confidence: 1.000

Max Lift: 2.963

Top Rules (by Lift)

Rule	Conf	Lift
Education_Category_High School|Experience_Category_Entry → Salary_Category_Low	1.00	2.96
Education_Category_High School|Job_Group_Marketing_Sales → Salary_Category_Low	1.00	2.96
Experience_Category_Entry|Job_Group_Marketing_Sales → Salary_Category_Low	1.00	2.96
Education_Category_High School|Experience_Category_Entry|Job_Group_Marketing_Sales → Salary_Category_Low	1.00	2.96
Experience_Category_Executive|Job_Group_Tech → Salary_Category_High	0.97	2.92
Experience_Category_Executive → Salary_Category_High	0.97	2.91
Education_Category_High School → Salary_Category_Low	0.97	2.87
Education_Category_PhD|Experience_Category_Senior|Job_Group_Tech → Salary_Category_High	0.92	2.76

Interpretation: Rules with lift greater than 1 indicate meaningful associations between career attributes and salary outcomes. Higher confidence reflects rule reliability, while lift captures the strength of the relationship beyond random chance.

Resume NLP Module (Feature Extraction System)

This module processes unstructured resume text and converts it into structured features used by the machine learning model. Unlike predictive models, this is a rule-based NLP system designed for efficient and interpretable feature extraction.

System Components

- Text Extraction: pdfplumber (PDF parsing)
- Text Cleaning: Regex preprocessing
- Experience Extraction: Regex-based pattern detection
- Education Detection: Rule-based keyword matching
- Skill Detection: spaCy PhraseMatcher
- Job Title Detection: Phrase matching with alias mapping
- Country Detection: spaCy Named Entity Recognition
- Seniority Detection: Derived from experience and job title

Processing Pipeline

PDF Resume → Text Extraction → Cleaning → NLP Processing → Feature Extraction → Structured Input → Machine Learning Model

Design Rationale

- No labeled resume dataset available for supervised NLP training
- Rule-based NLP ensures deterministic and interpretable outputs
- Faster processing compared to deep learning models
- Suitable for structured feature extraction (skills, experience, etc.)

Limitations

- Performance depends on resume formatting and keyword presence
- May miss implicit or uncommon skill expressions
- Does not perform deep semantic understanding